



DRIVER DROWSINESS DETECTION AND VEHICLE CONTROL SYSTEM USING PYTHON, ARUDINO AND L298N MOTOR DRIVER

A PROJECT REPORT

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BONAFIDE CERTIFICATE

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DRIVER DROWSINESS DETECTION AND VEHICLE CONTROL SYSTEM USING PYTHON ,ARUDINO AND L298N MOTOR DRIVER

ABSTRACT:

In recent years, road traffic accidents have become a major global concern, with driver fatigue being one of the main causes. Long driving hours, lack of sleep, stress, and physical exhaustion reduce a driver's alertness and concentration. As fatigue increases, reaction time slows down and decision-making ability decreases, increasing the chances of accidents. Even a short period of microsleep can lead to serious consequences such as injuries, loss of life, and property damage. Therefore, there is a need for an intelligent system that can monitor the driver and prevent such accidents. To address this problem, an automated Driver Drowsiness Detection and Vehicle Control System is proposed. The system monitors the driver's eye movements in real time using a webcam. The captured video is processed using computer vision techniques. Facial landmark detection is used to locate the eyes, and the Eye Aspect Ratio (EAR) is calculated to determine whether the eyes are open or closed. If the eyes remain closed for a certain duration, the system detects drowsiness.

Once drowsiness is detected, the system takes immediate safety actions. The processing is done using Python, which sends signals to the Arduino through serial communication. The Arduino controls the motor using the L298N motor driver. The system can reduce the vehicle speed or stop it to prevent accidents. At the same time, a buzzer and LED are activated to alert the driver. This system provides a simple, cost-effective, and real-time solution to improve road safety.

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LIST OF ABBREVIATIONS:

S.NO	ABBREVIATION	EXPANSION
1	EAR	Eye Aspect Ratio
2	DFD	Data Flow Diagram
3	UML	Unified Modeling Language
4	IoT	Internet of Things
5	AI	Artificial Intelligence
6	CPU	Central Processing Unit
7	USB	Universal Serial Bus

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CHAPTER 1

INTRODUCTION

1.1 INTRODUCTION:

In the modern world, transportation plays a vital role in daily life. With the increasing number of vehicles on the road, road safety has become a major concern. One of the most significant causes of road accidents worldwide is driver drowsiness. Long driving hours, lack of sleep, stress, and continuous concentration reduce the alertness level of drivers. When a driver becomes drowsy, reaction time decreases and decision-making ability becomes slow, which may result in severe accidents.

Traditional safety systems focus mainly on vehicle mechanics such as braking systems and airbag deployment. However, very few systems directly monitor the physical condition of the driver. Manual awareness alone is not sufficient to prevent fatigue-related accidents. Therefore, there is a strong need for an intelligent system that continuously monitors the driver's condition and takes preventive action automatically.

The proposed Driver Drowsiness Detection and Vehicle Control System uses computer vision techniques to monitor the driver's eye movements in real time. A webcam captures the driver's face continuously. The system detects facial landmarks and calculates the Eye Aspect Ratio (EAR) to determine whether the eyes are open or closed. If the eyes remain closed for a predefined time duration, the system detects drowsiness.

Once drowsiness is detected, the system sends a signal from Python to an embedded system using serial communication. The embedded controller used in this project is the Arduino Uno. The Arduino controls the DC motor using the L298N Motor Driver Module. When an emergency condition occurs, the motor speed is reduced or stopped completely. At the same time, a buzzer and LED are activated to alert the driver.

This project demonstrates an effective integration of software and hardware technologies to enhance vehicle safety. The system is cost-effective, reliable, and suitable for real-time implementation.

1.2 PROJECT OBJECTIVE:

The primary objective of this project is to design and develop a real-time driver drowsiness detection system that automatically controls vehicle movement and reduces accident risk.

The detailed objectives of the system are as follows:

- To capture live video of the driver using a webcam.
- To detect face and extract eye landmarks using Dlib.
- To calculate Eye Aspect Ratio for fatigue detection.
- To identify prolonged eye closure indicating drowsiness.
- To send control signals to Arduino through serial communication.
- To regulate motor speed using L298N motor driver.
- To activate alert mechanisms such as buzzer and LED.
- To design a low-cost prototype suitable for safety applications.

SCOPE:

The scope of the Driver Drowsiness Detection System extends to improving road safety by minimizing fatigue-related accidents. The system can be implemented in various types of vehicles such as cars, trucks, buses, and commercial transport vehicles.

The project focuses on:

- Real-time monitoring.
- Automated safety response.
- Embedded system integration.
- Software-hardware communication.

In future developments, the system can be integrated with actual vehicle control units (ECU) for practical deployment.

1.3 EXISTING SYSTEM AND PROPOSED SYSTEM:

Existing System	Proposed System
In the existing system, driver fatigue detection is not automated in most vehicles. Safety mechanisms depend mainly on manual driver awareness. Some advanced vehicles use basic alarm systems, but they do not automatically control the real vehicle movement.	The proposed system introduces an intelligent and automated solution for detecting driver drowsiness and controlling vehicle speed.
Techniques: The Vehicle behavior monitoring, physiological sensors, or deep learning techniques.	Techniques: Lightweight computer vision-based EAR technique combined with Arduino control, making it cost-effective, real-time, and suitable for embedded implementation.
Demerits: • No continuous monitoring of eye • Movement. • No automatic speed reduction • mechanism • Dependence on driver response. • High probability of human error.	Merits: • The system provides real-time monitoring of the driver's eye movements. • It detects drowsiness at an early stage using Eye Aspect Ratio (EAR). • Automatic vehicle speed control reduces accident risk. • The Arduino Uno ensures reliable motor control. • The L298N Motor Driver Module enables efficient speed regulation. • Immediate alert is provided through buzzer and LED indication

1.4 LITERATURE SURVEY:

1.4.1 REAL TIME BLINK DETECTION:

TITLE : real-time blink detection using facial landmarks

AUTHOR : Soukupová & Čech

YEAR : 2016

DESCRIPTION:

Tereza Soukupová and Jan Čech introduced the Eye Aspect Ratio (EAR) method for real-time blink detection using facial landmark points. This method calculates a ratio based on the vertical and horizontal distances between specific eye landmarks. Because the human eye maintains a nearly constant horizontal width while the vertical distance decreases during blinking or eye closure, the EAR value drops significantly when the eyes are closed.

In practical implementation, the system continuously computes the EAR value for each video frame. When the EAR falls below a predefined threshold and remains low for a certain number of consecutive frames, the system identifies that the eyes are closed for an abnormal duration. This condition is considered a strong indicator of drowsiness rather than a normal blink.

The EAR method is simple, computationally efficient, and highly suitable for real-time applications. It does not require complex training datasets or heavy processing power, making it ideal for embedded and low-cost systems. Due to its reliability and speed, the proposed driver drowsiness detection system directly implements the EAR method to accurately monitor eye closure and detect driver fatigue in real time.

The EAR approach also works efficiently with standard webcams and low-cost cameras, which makes it suitable for affordable driver monitoring systems. Because it requires minimal computational resources, the algorithm can run smoothly on systems with limited processing power.

Furthermore, the EAR-based detection technique can be easily integrated with other safety mechanisms such as alarm systems, vehicle control modules, and IoT-based alert systems. This integration improves the overall reliability of driver safety systems by ensuring that timely warnings are provided whenever abnormal eye closure patterns are detected.

Due to these advantages, the EAR method has become one of the most widely adopted techniques in driver drowsiness detection research and practical implementations.

1.4.2 CNN BASED APPROACH:

TITLE: (CNN)-based approach for driver fatigue detection.

AUTHOR : Zhang et al.

YEAR : 2017

DESCRIPTION:

Zhang and colleagues proposed a driver fatigue detection system based on Convolutional Neural Networks (CNNs), a deep learning technique capable of automatically extracting complex features from facial images. Unlike traditional methods that rely on manually defined features such as eye aspect ratio or blink rate, CNN models learn important patterns directly from large datasets of driver images. This allows the system to detect subtle signs of fatigue, including changes in eye appearance, facial expressions, and head posture.

One of the major advantages of CNN-based approaches is their high accuracy, especially under challenging conditions such as varying lighting, different camera angles, and partial occlusions. Deep learning models can generalize well when trained on diverse datasets, making them suitable for real-world applications.

However, deep learning methods require large labeled datasets for training and significant computational resources, including powerful GPUs. This increases system cost and complexity, making them less suitable for low-cost embedded implementations. Therefore, for real-time and cost-effective systems, lightweight techniques such as the Eye Aspect Ratio (EAR) method remain practical and efficient, especially when integrated with microcontrollers like Arduino for vehicle control applications.

However, CNN systems require a large amount of labeled training data to achieve high accuracy. Collecting such datasets can be time-consuming and expensive. In addition, deep learning models require high computational power for training and real-time processing, which may not be feasible for embedded hardware platforms.

Despite these challenges, CNN-based fatigue detection systems continue to be widely researched because of their ability to achieve high accuracy in complex driving environments.

1.4.3 MACHINE LEARNING BASED

TITLE: Machine Learning-Based Driver Drowsiness Detection System Using Facial Feature Extraction and Classification Technique

AUTHOR: Dasgupta et al.

YEAR: 2019

DESCRIPTION:

Dasgupta and his team developed a driver drowsiness detection system based on machine learning techniques combined with facial feature extraction. In their approach, important facial features such as eye state, blink frequency, and facial expressions were first extracted from video frames. These features were then fed into classification algorithms to determine whether the driver was alert or drowsy.

Unlike traditional threshold-based methods that rely on fixed values (such as a predefined EAR limit), machine learning models can learn patterns from training data and make more adaptive decisions. This improves detection accuracy, especially in real-world conditions where lighting, facial structure, and driver behavior may vary.

The study demonstrated that integrating machine learning algorithms enhances system performance, robustness, and adaptability. It also highlighted that intelligent classification models can reduce false alarms and provide more reliable fatigue detection compared to simple rule-based systems.

Another important advantage of machine learning methods is their adaptability. Unlike rule-based systems that depend on fixed thresholds, machine learning models can adjust to different driving conditions, lighting environments, and driver behaviors.

The research by Dasgupta et al. demonstrated that combining facial feature extraction with machine learning classification techniques significantly improves the reliability of driver monitoring systems. This approach helps reduce false detections and ensures more accurate identification of driver fatigue in real-world scenarios.

CHAPTER 2

PROJECT DESCRIPTION

2.1 PROBLEM DEFINITION:

Driver drowsiness is one of the leading causes of fatal road accidents globally. Many drivers are unaware when they begin to lose alertness. Even a few seconds of microsleep can lead to serious accidents. Therefore, a reliable and automated system is required to detect early signs of drowsiness and take corrective action.

The challenge is to design a system that:

- Works in real time.
- Is cost-effective.
- Is easy to implement.
- Provides automatic control response.

This project addresses these challenges using computer vision and embedded systems.

2.2 METHODOLOGIES:

The system follows the below methodology:

- a. Capture real-time video using webcam.
- b. Detect face using Dlib library.
- c. Extract eye landmarks.
- d. Calculate Eye Aspect Ratio (EAR).
- e. Compare EAR with threshold value.
- f. If threshold crossed:
 - i. Send signal to Arduino.
- g. Arduino controls motor using L298N driver.
- h. Buzzer and LED alert activated.

2.3 MODULES:

- **Video Capture Module:** Captures real-time video of the driver using a webcam for continuous monitoring.
- **Face Detection Module:** Detects the driver's face from each video frame using computer vision techniques.
- **Eye Landmark Detection Module:** Identifies eye region points using facial landmark detection for further analysis.
- **EAR Calculation Module:** Calculates the Eye Aspect Ratio (EAR) to determine whether the eyes are open or closed.
- **Drowsiness Decision Module:** Compares EAR value with threshold and confirms drowsiness if eyes remain closed for a specific duration.
- **Serial Communication Module:** Sends control signals from Python to the Arduino Uno through USB communication.
- **Motor Control Module:** Controls the DC motor speed using the L298N Motor Driver Module based on received signals.
- **Alert Module:** Activates buzzer and LED to warn the driver during drowsiness detection.

2.4 ALGORITHM USED:

2.4.1 EYE ASPECT RATIO ALGORITHM:

The Eye Aspect Ratio is calculated using Euclidean distance between eye landmark points.

Formula:

$$EAR = (\|p2-p6\| + \|p3-p5\|) / (2 \times \|p1-p4\|)$$

Where:

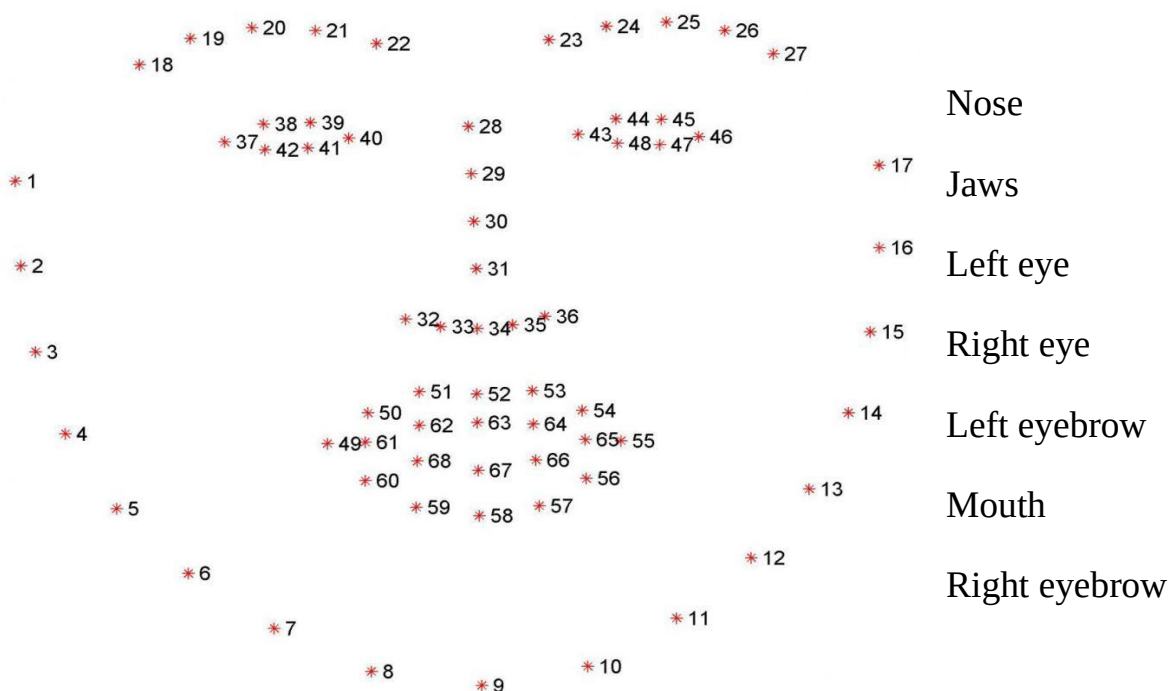
- p1 to p6 are eye landmark points.

If EAR remains below threshold (e.g., 0.25) for 20 consecutive frames, the driver is considered drowsy.

2.4.2 DLIB'S FACIAL LANDMARK DETECTOR:

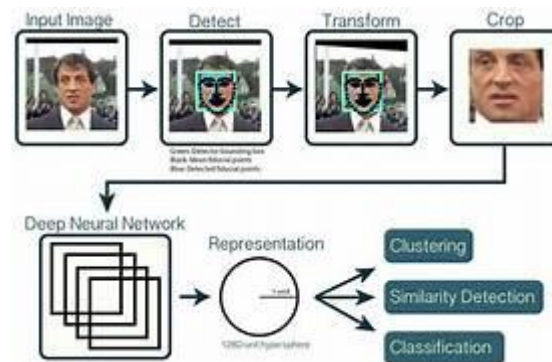
The pre-trained facial landmark detector inside the dlib library is used to estimate the location of 68 (x, y)-coordinates that map to facial structures on the face.

Facial landmarks are used for localizing and representing salient regions or facial parts of the person's face, such as:



2.4.3 OPENCV:

OpenCV is an open-source computer vision and image processing library developed by Intel in 2000. It provides various tools and functions for real-time image and video processing applications.



2.4.4 HAAR CASCADE ALGORITHM :

- Used to detect the driver's face from real-time video.
- Based on Haar-like features and a cascade classifier.
- Quickly identifies the face region in each frame.

2.4.5 THRESHOLD BASED ALGORITHM:

- Compares EAR value with a fixed threshold.
- If condition is satisfied for a specific time, system triggers:
- Buzzer alert
- LED indication
- Vehicle speed control using Arduino

CHAPTER 3

REQUIREMENTS

3.1 GENERAL:

These are the requirements for doing the project. Without using these tools and software's we can't do the project. So we have two requirements to do the project. They are

1. Hardware Requirements.
2. Software Requirements.

3.2 HARDWARE REQUIREMENTS:

The hardware requirements define the physical components necessary to implement the real-time driver drowsiness detection system. These requirements provide a foundation for system design and ensure reliable performance during continuous monitoring and alert generation.

PROCESSOR	: Intel Core i3 / i5 or higher (Minimum Pentium IV 2.6 GHz)
RAM	: 4 GB (Minimum)
MONITOR	: 15" Color Monitor
HARD DISK	: 40 GB or Higher
CAMERA	: USB Webcam
MICRO CONTROLLER	: Arduino Uno
MOTOR DRIVER	: L298N Motor Driver Module
ACTUATOR	: DC Motor
ALERT DEVICES	: Buzzer , LED
POWER SUPPLY	: 12V DC Supply / Battery

3.3 SOFTWARE REQUIREMENTS:

The software requirements specify the tools and platforms used to develop, execute, and test the drowsiness detection system. These requirements focus on system functionality rather than implementation details and help in planning, development, and performance evaluation.

PROGRAMMING LANGUAGE : Python 3.7 +

LIBRARIES USED : OpenCV, dlib, imutils, NumPy, SciPy, PySerial

MICROCONTROLLER PROGRAM : Arduino IDE

OPERATING SYSTEM : Windows 10 / Windows 11

DEVELOPMENT ENVIRONMENT : VS Code / Python IDLE

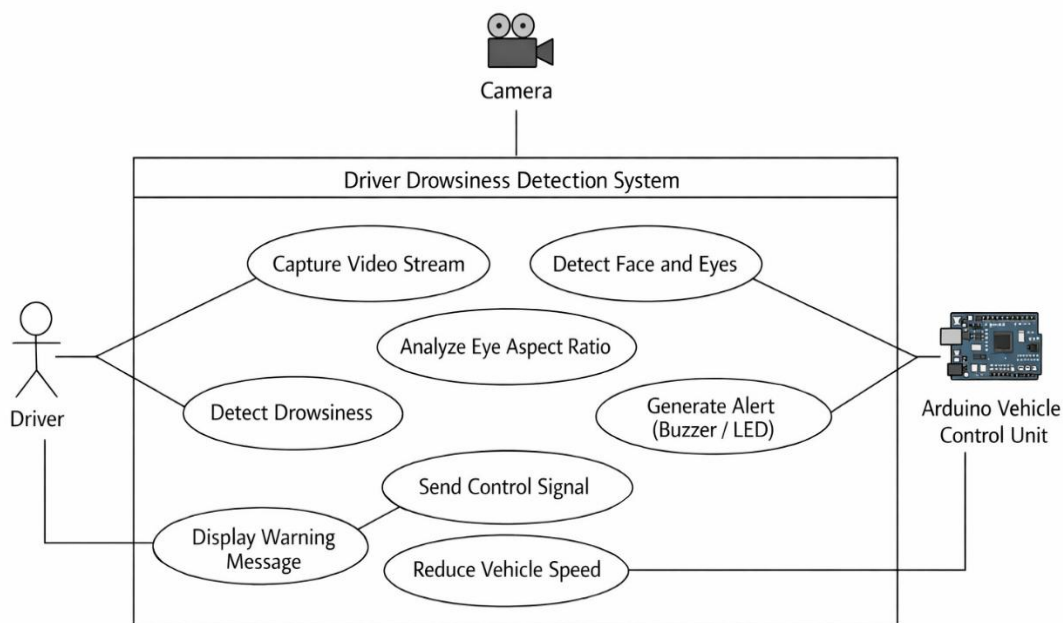
CHAPTER 4

DESIGN ENGINEERING

4.1 GENERAL:

Design engineering explains the overall design of the proposed driver drowsiness detection system. It describes how different components such as camera input, image processing, drowsiness detection algorithm and alert system work together. Various UML diagrams like usecase , class , sequence ,activity and system architecture are used to represent the structure and working flow of the system. These diagrams help to understand the interaction between system components and the overall functionality of the project.

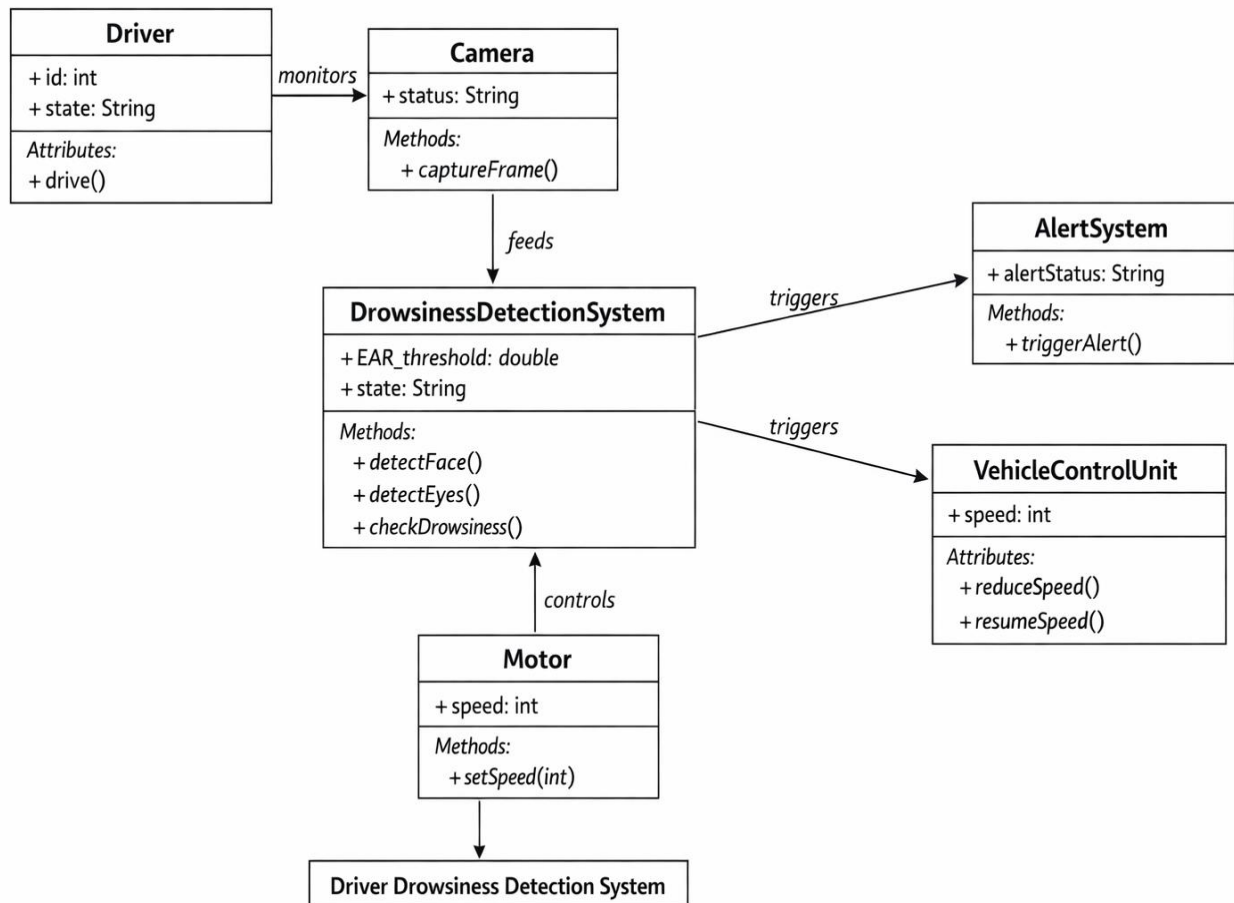
4.1.1 USE CASE DIAGRAM:



EXPLANATION:

The use case diagram represents the functional behavior of the Driver Drowsiness Detection System. It shows how the driver, camera, and vehicle control unit interact with the system. The camera captures the driver's face, the system analyzes eye movements to detect drowsiness, and alerts are generated. If drowsiness is detected, the system sends signals to reduce vehicle speed, ensuring driver safety.

4.1.2 CLASS DIAGRAM :

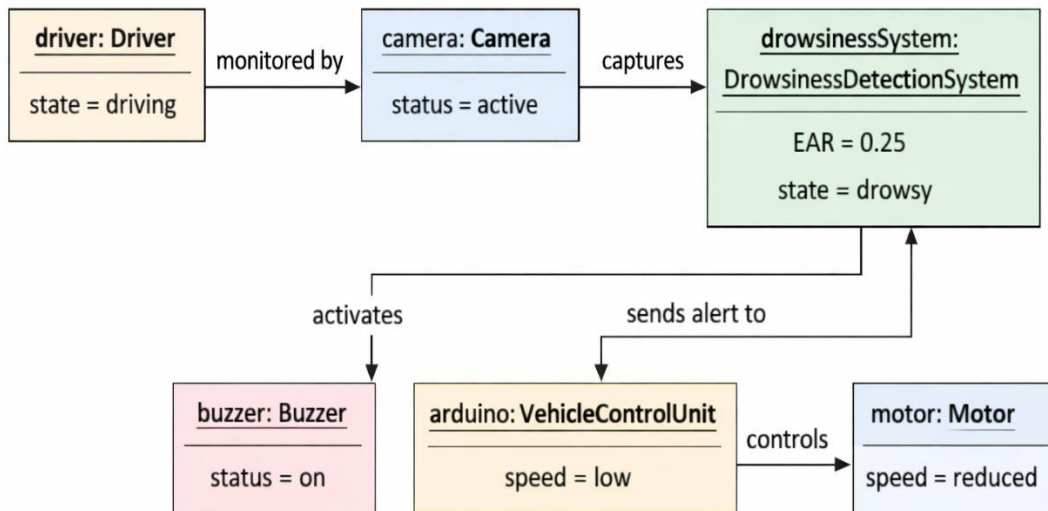


EXPLANATION:

The class diagram represents the structural design of the Driver Drowsiness Detection System. It shows the main classes such as Driver, Camera, Drowsiness Detection System, Alert System, and Vehicle Control Unit along with their attributes and methods. The diagram explains how the camera captures data, the system detects drowsiness, generates alerts, and controls vehicle speed, ensuring safe operation.

4.1.3 UML OBJECT DIAGRAM:

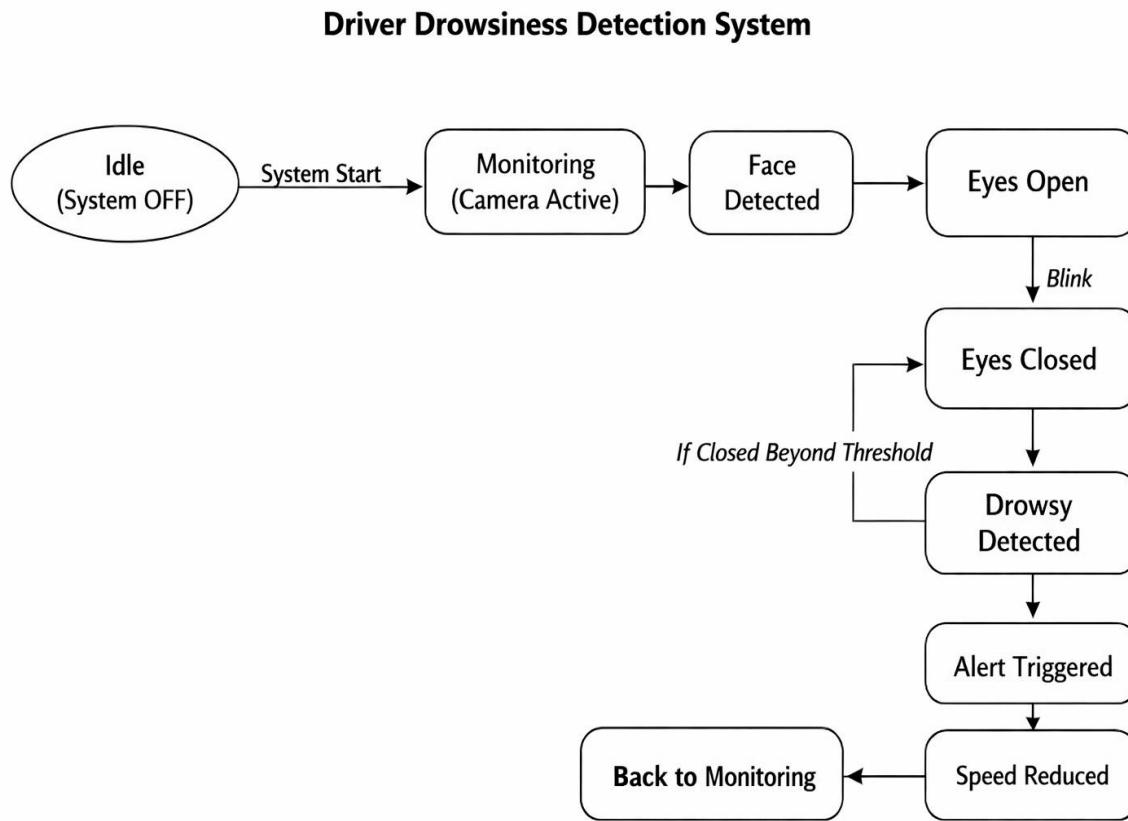
UML Object Diagram – Driver Drowsiness Detection System



EXPLANATION:

The UML object diagram represents the real-time instances of objects in the Driver Drowsiness Detection System. It shows how objects such as the driver, camera, drowsiness detection system, buzzer, Arduino unit, and motor interact at a specific moment. When drowsiness is detected, alerts are activated and the vehicle speed is reduced, ensuring driver safety.

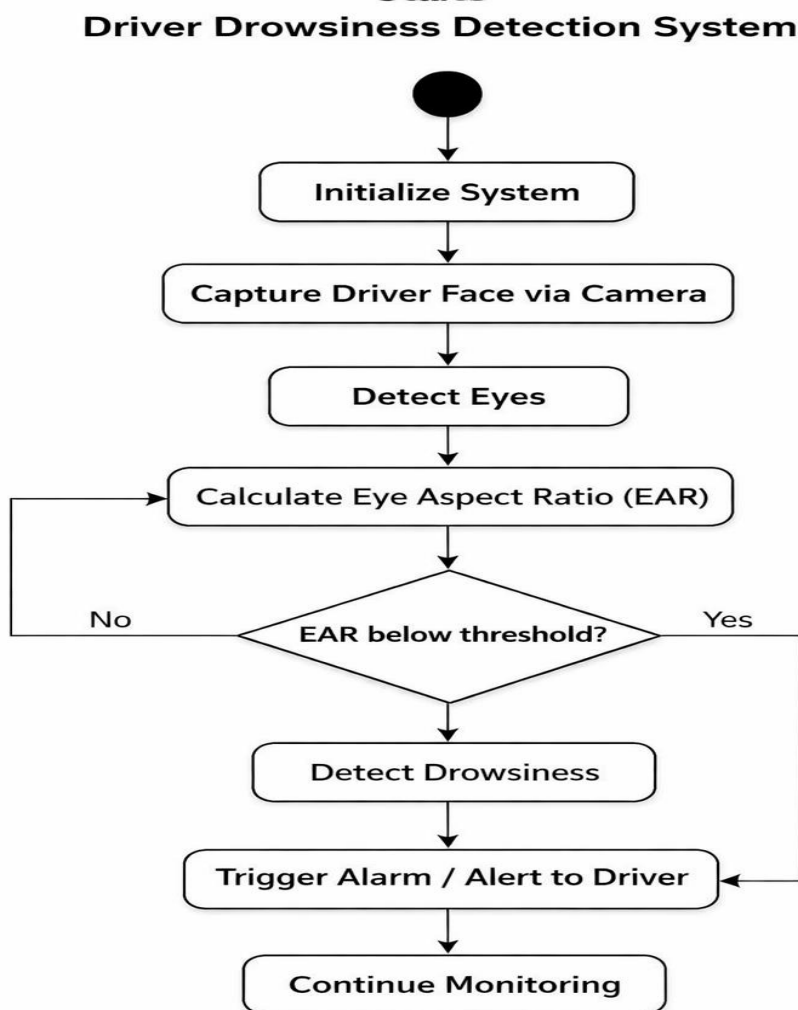
4.1.4 STATE DIAGRAM:



EXPLANATION:

The state diagram represents the different states of the Driver Drowsiness Detection System during operation. The system starts in an idle state, then moves to monitoring when activated. It detects the driver's face and eye status continuously. If the eyes remain closed beyond a threshold, drowsiness is detected, alerts are triggered, and vehicle speed is reduced. After action, the system returns to monitoring mode.

4.1.5 ACTIVITY DIAGRAM :

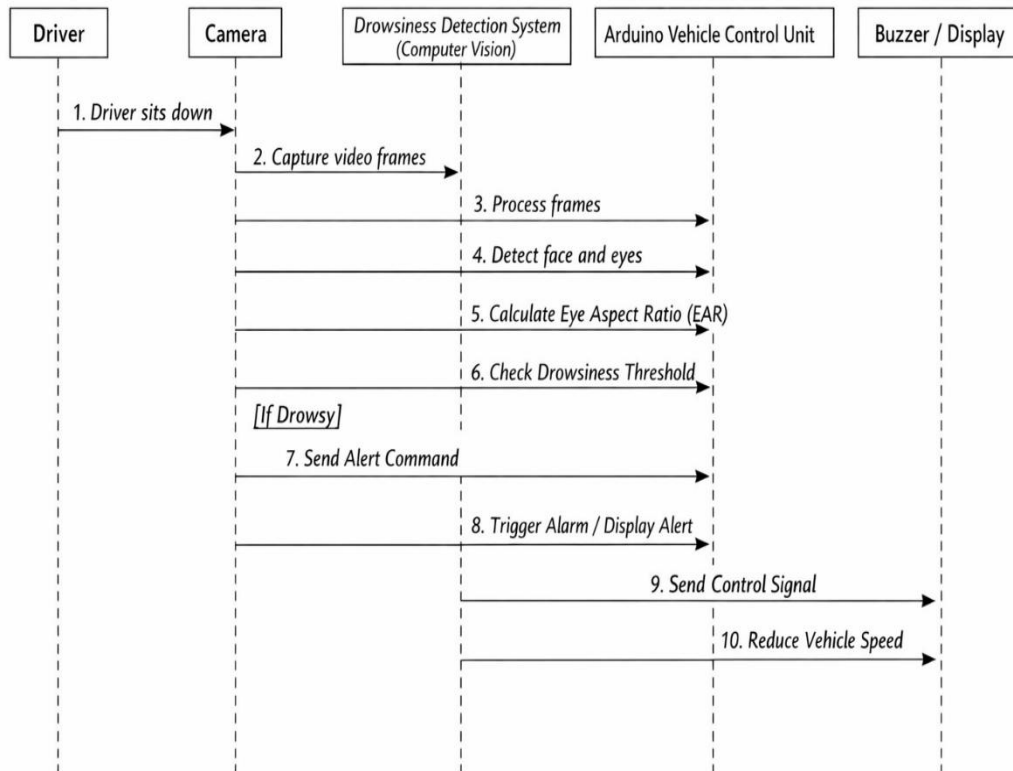


EXPLANATION:

The activity diagram represents the workflow of the driver drowsiness detection system. The process starts by initializing the system and also activating the camera to capture the driver's face. The system then detects the driver's eyes and calculates the Eye Aspect Ratio (EAR) using image processing techniques. If the EAR value falls below the predefined threshold, the system identifies that the driver is drowsy. Once drowsiness is detected, an alert or alarm is triggered to warn the driver. If the EAR value is normal, the system itself continues monitoring the driver continuously to ensure safety.

4.1.6 SEQUENCE DIAGRAM:

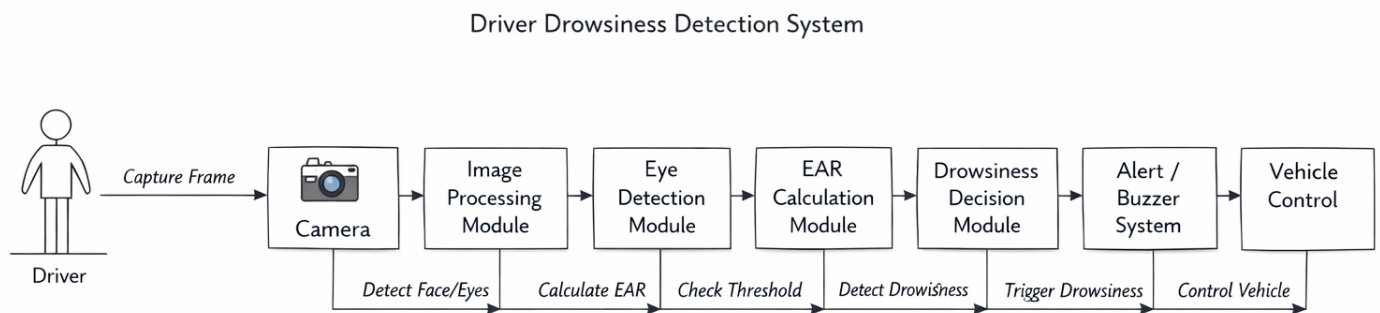
Driver Drowsiness Detection System



EXPLANATION:

The sequence diagram illustrates the step-by-step interaction between the driver, camera, drowsiness detection system, and vehicle control unit. The camera continuously captures the driver's facial video, which is processed to detect eye movements and drowsiness. When fatigue is detected, the system generates an alert and sends control signals to reduce vehicle speed, ensuring driver safety.

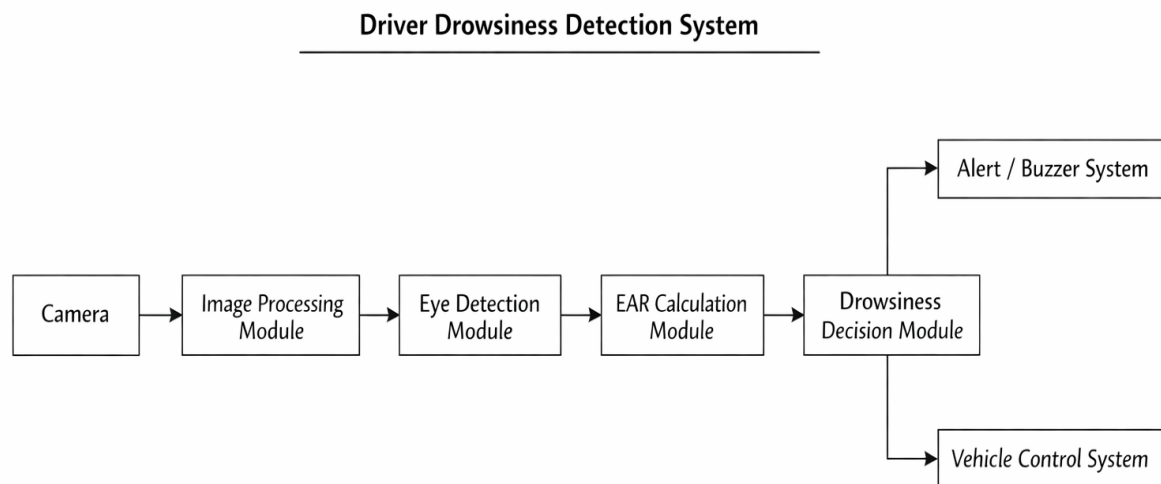
4.1.7 COLLABORATION DIAGRAM :



EXPLANATION:

The collaboration diagram shows how different components of the system interact with each other to detect driver drowsiness. The camera captures the driver's face and sends the image to the image processing module. The eye detection module identifies the driver's eyes and calculates the Eye Aspect Ratio (EAR). If the EAR value falls below the threshold, the drowsiness decision module detects driver fatigue. Then the alert system activates a buzzer to warn the driver and may also control the vehicle for safety.

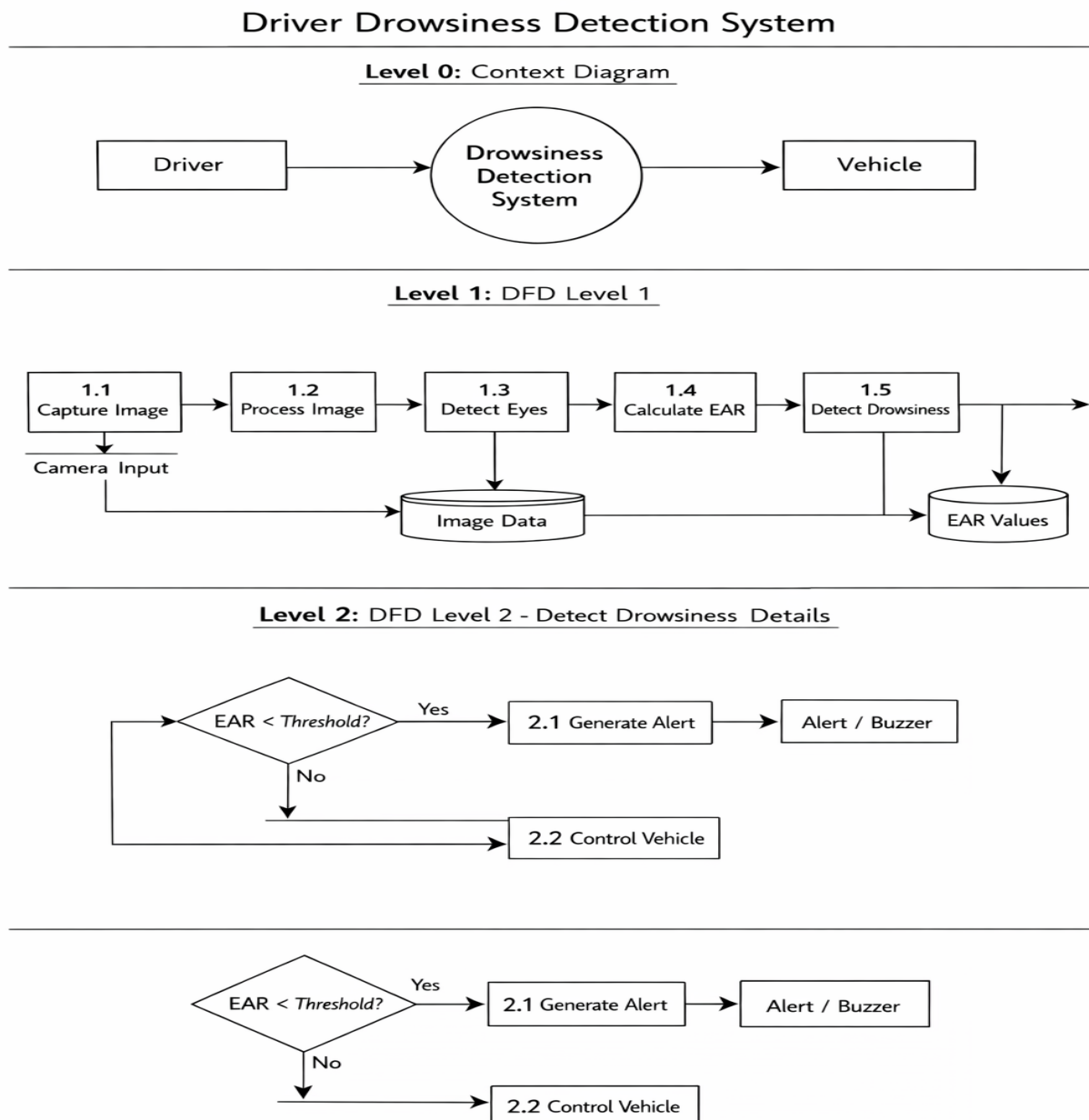
4.1.8 COMPONENT DIAGRAM :



EXPLANATION:

The component diagram shows the main modules of the driver drowsiness detection system and how they are connected. The camera captures the driver's image and sends it to the image processing module. The eye detection module then identifies the driver's eyes and sends the data to the EAR calculation module to compute the Eye Aspect Ratio. The drowsiness decision module checks whether the driver is drowsy based on the EAR value. If drowsiness is detected, the system activates the alert or buzzer system to warn the driver and can also send signals to the vehicle control system for safety.

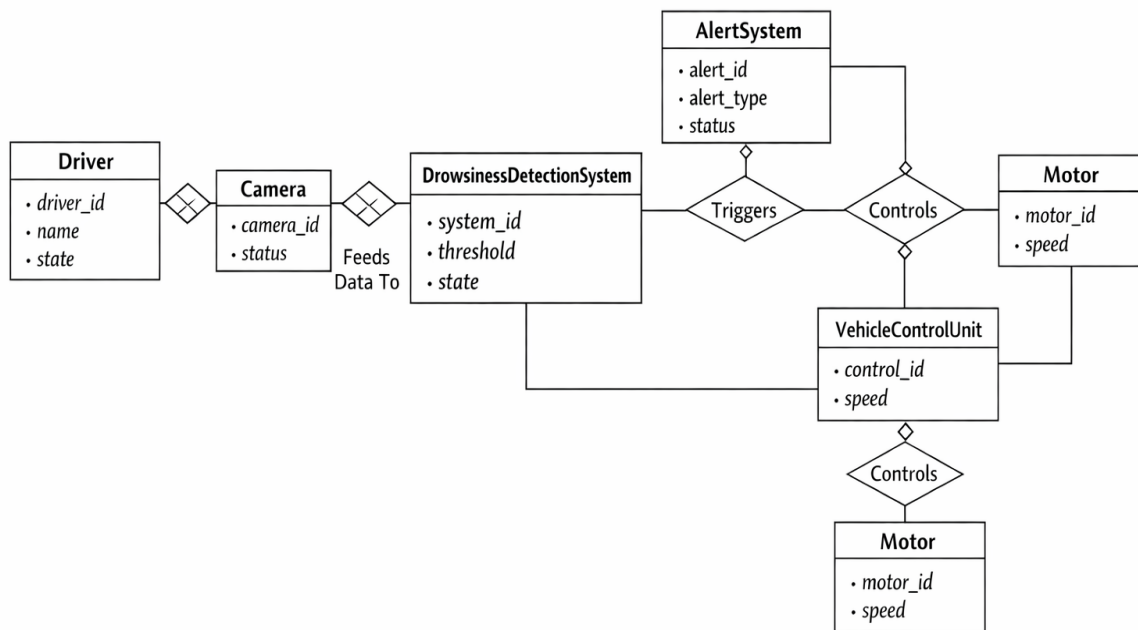
4.1.9 DATA FLOW DIAGRAM:



EXPLANATION:

The data flow diagram shows how data moves through the Driver Drowsiness Detection System. It explains how the camera captures the driver's facial data, which is processed by the drowsiness detection module. Based on the analysis, alert signals are generated and control commands are sent to the vehicle control unit to ensure safe driving.

4.1.10 E-R DIAGRAM:

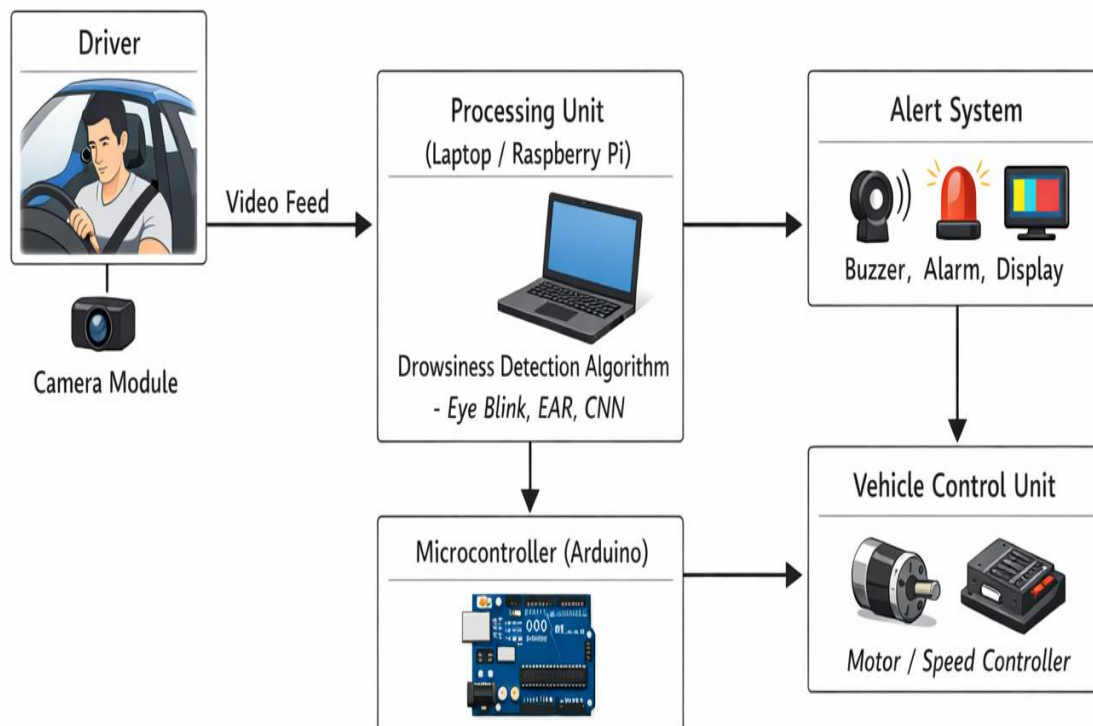


EXPLANATION:

The ER diagram represents the data structure of the Driver Drowsiness Detection System. It shows the main entities such as Driver, Camera, Drowsiness Detection System, Alert System, Vehicle Control Unit, and Motor along with their attributes. The diagram explains how camera data is processed to detect drowsiness, trigger alerts, and control vehicle speed to ensure driver safety.

4.1.11 SYSTEM ARCHITECHTURE:

Real-Time Driver Drowsiness Detection System



EXPLANATION:

The system architecture shows the overall working of the Driver Drowsiness Detection System. A camera captures the driver's facial video and sends it to the processing unit, where drowsiness is detected using image processing algorithms. When drowsiness is identified, alerts are generated, and control signals are sent through the Arduino to the vehicle control unit to reduce speed and improve safety.

CHAPTER 5

IMPLEMENTATION

5.1 GENERAL:

This chapter describes the implementation of the Real-Time Driver Drowsiness Detection System. It explains how the system detects driver fatigue using image processing and controls alerts and vehicle speed for safety.

5.2 SOURCE CODE FOR DROWSINESS :

```
import cv2
import dlib
import imutils
import serial
import time
from scipy.spatial import distance as dist
from imutils import face_utils

# ----- SETTINGS ----- #

EAR_THRESHOLD = 0.25
MILD_FRAMES = 20
SEVERE_FRAMES = 40
EXTREME_FRAMES = 60

COUNTER = 0

# ----- CONNECT TO ARDUINO ----- #

arduino = serial.Serial('COM9', 9600) # Change COM port
time.sleep(2)

def eye_aspect_ratio(eye):
    A = dist.euclidean(eye[1], eye[5])
    B = dist.euclidean(eye[2], eye[4])
```



```

C = dist.euclidean(eye[0], eye[3])
return (A + B) / (2.0 * C)

detector = dlib.get_frontal_face_detector()
predictor = dlib.shape_predictor("shape_predictor_68_face_landmarks.dat")

(lStart, lEnd) = face_utils.FACIAL_LANDMARKS_IDXS["left_eye"]
(rStart, rEnd) = face_utils.FACIAL_LANDMARKS_IDXS["right_eye"]

cap = cv2.VideoCapture(0)
print("System Started...")

while True:
    ret, frame = cap.read()
    if not ret:
        break

    frame = imutils.resize(frame, width=600)
    gray = cv2.cvtColor(frame, cv2.COLOR_BGR2GRAY)

    rects = detector(gray, 0)

    for rect in rects:
        shape = predictor(gray, rect)
        shape = face_utils.shape_to_np(shape)

        leftEye = shape[lStart:lEnd]
        rightEye = shape[rStart:rEnd]

        leftEAR = eye_aspect_ratio(leftEye)
        rightEAR = eye_aspect_ratio(rightEye)
        EAR = (leftEAR + rightEAR) / 2.0

        if EAR < EAR_THRESHOLD:
            COUNTER += 1

        if COUNTER >= EXTREME_FRAMES:
            cv2.putText(frame, "EXTREME - VEHICLE STOPPED", (20,80),
                cv2.FONT_HERSHEY_SIMPLEX, 0.7, (0,0,255), 3)
            arduino.write(b'E')

```

```

elif COUNTER >= SEVERE_FRAMES:
    cv2.putText(frame, "SEVERE DROWSINESS", (20,60),
                 cv2.FONT_HERSHEY_SIMPLEX, 0.7, (0,0,255), 2)
    arduino.write(b'S')

elif COUNTER >= MILD_FRAMES:
    cv2.putText(frame, "MILD DROWSINESS", (20,40),
                 cv2.FONT_HERSHEY_SIMPLEX, 0.7, (0,255,255), 2)
    arduino.write(b'M')

else:
    COUNTER = 0
    arduino.write(b'N')

cv2.putText(frame, f"EAR: {EAR:.2f}", (450,30),
             cv2.FONT_HERSHEY_SIMPLEX, 0.7, (255,255,255), 2)

cv2.imshow("Driver Monitoring", frame)

if cv2.waitKey(1) & 0xFF == ord("q"):
    break

cap.release()
cv2.destroyAllWindows()
arduino.close()

```

ARDUINO CODE:

```

int motorPin1 = 8;
int motorPin2 = 9;
int enablePin = 10;
int buzzer = 6;
int led = 7;
void setup() {
    pinMode(motorPin1, OUTPUT);
    pinMode(motorPin2, OUTPUT);
    pinMode(enablePin, OUTPUT);
    pinMode(buzzer, OUTPUT);
    pinMode(led, OUTPUT);
}

```

```

Serial.begin(9600);

// Default motor direction
digitalWrite(motorPin1, HIGH);
digitalWrite(motorPin2, LOW);
analogWrite(enablePin, 200);
}
void loop() {

  if (Serial.available() > 0) {
    char state = Serial.read();

    if (state == 'N') { // Normal
      digitalWrite(motorPin1, HIGH);
      digitalWrite(motorPin2, LOW);
      analogWrite(enablePin, 200);
      digitalWrite(buzzer, LOW);
      digitalWrite(led, LOW);
    }

    if (state == 'S') { // Slow
      analogWrite(enablePin, 100);
      digitalWrite(led, HIGH);
    }

    if (state == 'E') { // Emergency Stop
      digitalWrite(motorPin1, LOW);
      digitalWrite(motorPin2, LOW);
      digitalWrite(enablePin, LOW);
      digitalWrite(led, HIGH);
      digitalWrite(buzzer, HIGH);
    }
  }
}

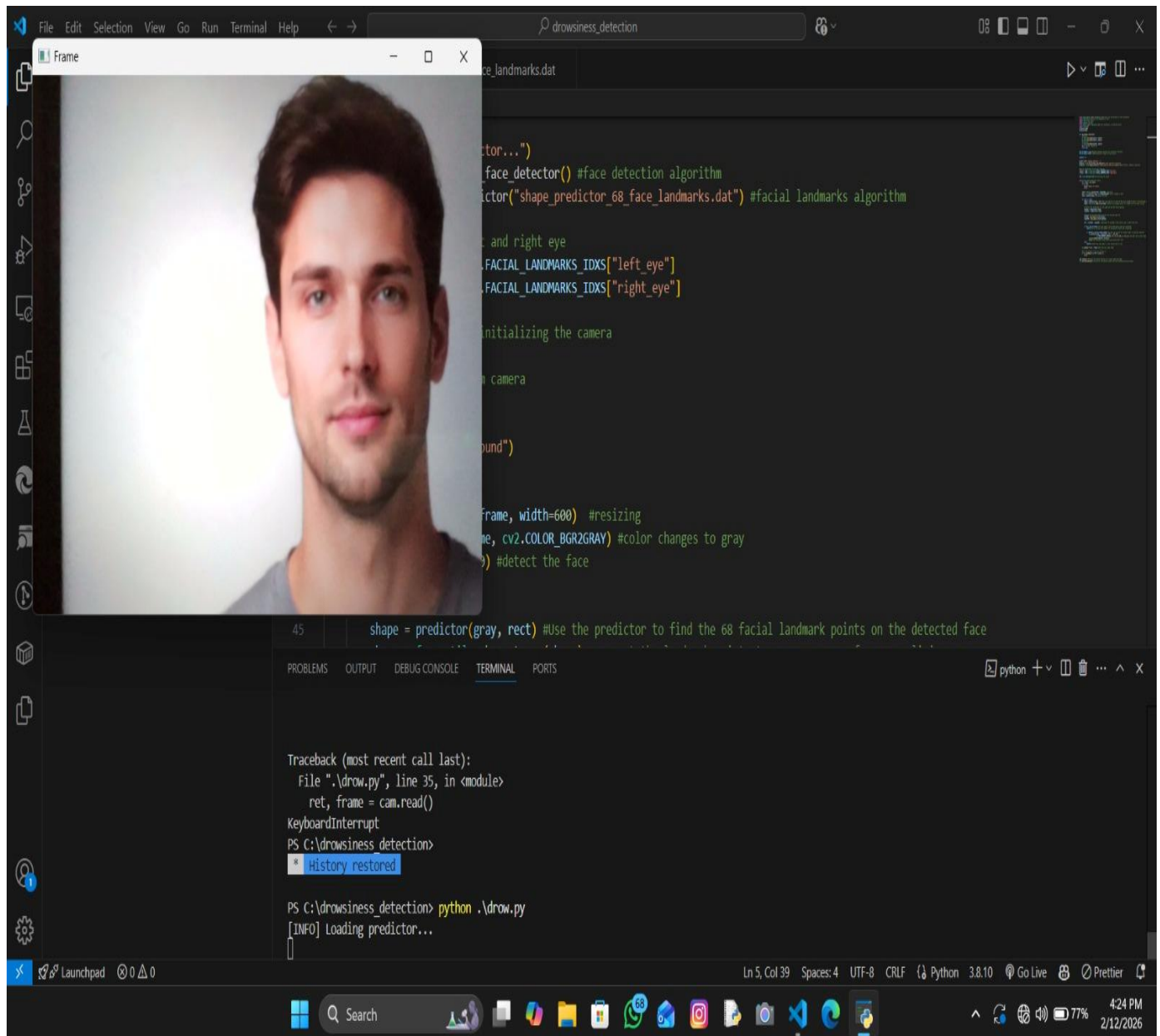
```

CHAPTER 6

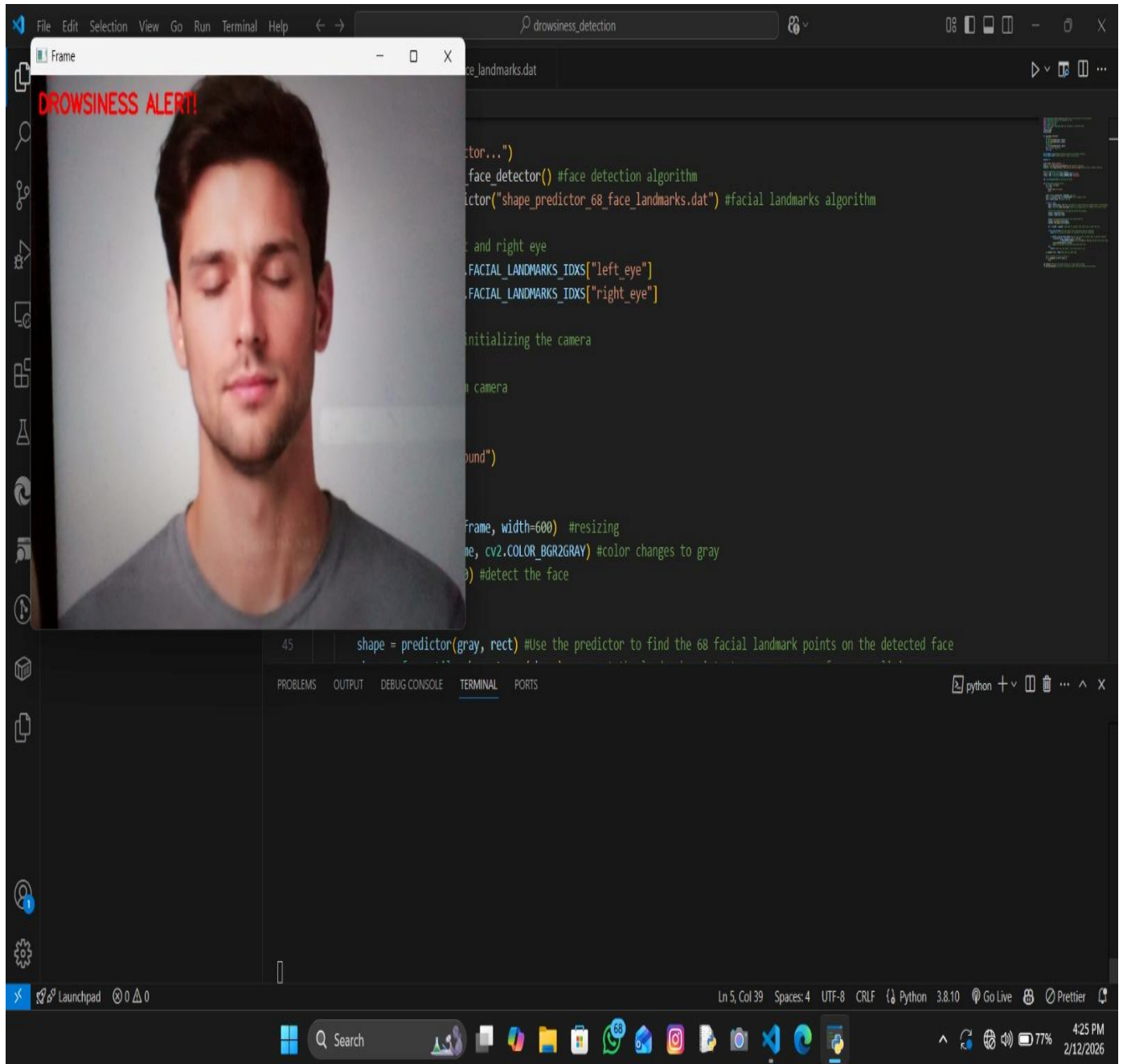
SNAPSHOTS

6.1 VARIOUS SNAPSHOTS:

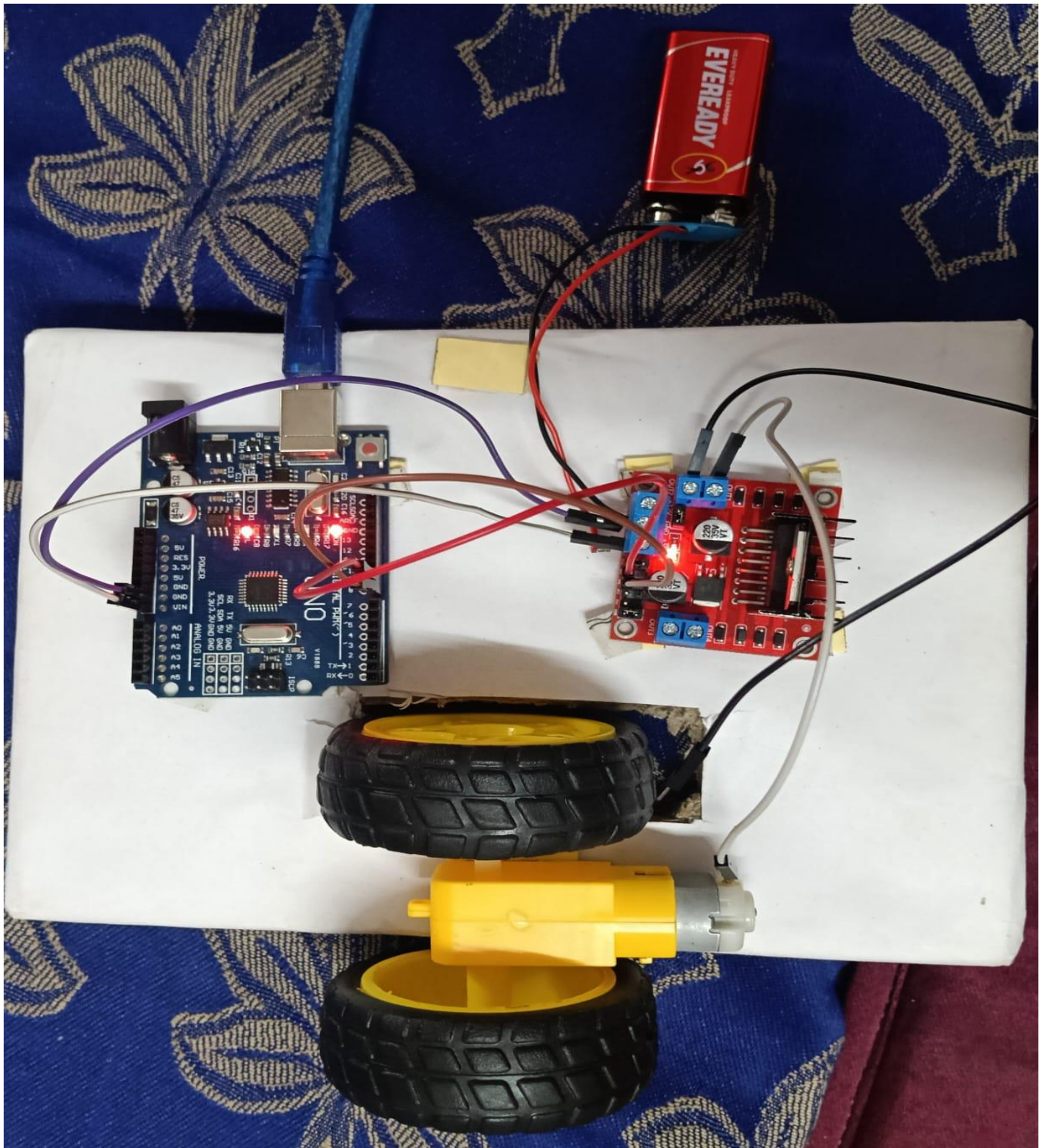
6.1.1 EYES OPEN DETECTION OUTPUT:



6.1.2 EYES CLOSED DETECTION OUTPUT:



6.1.3 ARDUINO VEHICLE CONTROL OUTPUT:



CHAPTER 7

REFERENCES

7.1 CONCLUSION:

The Real-Time Driver Drowsiness Detection with Automated Vehicle Speed Control system is designed to enhance road safety by continuously monitoring the driver's alertness level. The system uses a camera to capture facial features and applies image processing techniques such as eye blink detection and Eye Aspect Ratio (EAR) analysis to identify signs of drowsiness in real time. By analyzing the driver's eye movements frame by frame, the system can accurately determine whether the driver is alert or experiencing fatigue.

When drowsiness is detected, the system immediately activates alert mechanisms such as a buzzer and visual warning to wake the driver. These alerts are designed to quickly grab the driver's attention and prevent dangerous situations. If the driver does not respond within a short period, the Arduino-based vehicle control unit automatically reduces the vehicle speed or stops the motor completely to avoid possible accidents. This automatic intervention adds an extra layer of safety, especially in critical situations where the driver may lose control.

The system integrates both software and hardware components effectively. The image processing and detection tasks are performed using Python, while the Arduino microcontroller handles real-time control of the motor through the L298N driver module. This combination ensures smooth communication between detection and action, making the system reliable and efficient.

This project demonstrates how embedded systems, image processing, and automation can be combined to create an intelligent safety solution. It provides a cost-effective, reliable, and practical approach for reducing accidents caused by driver fatigue. Furthermore, the system can be enhanced in the future by incorporating advanced technologies such as deep learning, IoT-based alert systems, and real-time vehicle tracking, making it more suitable for modern automotive applications.

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<https://opencv.org/>

Used for image processing and computer vision tasks such as face and eye detection.

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Provides facial landmark detection used to locate eyes for EAR calculation.

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Soukupová, T., & Čech, J. (2016). Real-Time Eye Blink Detection using Facial Landmarks.

<https://vision.fe.uni-lj.si/cvww2016/proceedings/papers/05.pdf>

4.IEEE Xplore Digital Library

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Contains many research papers related to driver drowsiness detection systems.

5.Python Programming Language

<https://www.python.org/>

Used for implementing the detection algorithm and integrating computer vision modules.

6.National Highway Traffic Safety Administration

<https://www.nhtsa.gov/road-safety/drowsy-driving>

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7.2.2 BOOK REFERENCES :

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